THE SKEW HOOK FORMULA DOES NOT EXTEND TO SIMPLE-SPECTRUM HEAPS

JAIDEEP SAI PADHI

Abstract. The coloured posets arising from simple-spectrum torus fixed points [1, §5.4.2] are hook length posets but need not be d-complete. We show that the skew hook formula of Naruse–Okada [3, Thm. 1.2], read with its own hook function, does not extend to them. The obstruction is the hook function rather than the sum over excited diagrams, and it is already present at $F = \varnothing$. The witness has eight elements and the verification is finite: at most eight cases. Since the poset is a hook length poset, the $F = \varnothing$ identity does hold for it with the correct hook function, so only the recipe for h is at fault; we state the resulting question.

1. The question and the answer

Let Γ be a simply-laced Dynkin diagram with Weyl group W, and let $w \in W$ have reduced word $s_{i_1} \cdots s_{i_\ell}$. Its *heap* [1, Def. 5.2] is the poset on $\{1, \dots, \ell\}$ generated by the relations between positions carrying equal or adjacent colours, each element a coloured by i_a. For λ-minuscule w the heap is d-complete and the Peterson–Proctor formula [3, Thm. 1.1] evaluates its reverse-plane-partition series. Dinkins, Karpov and Krylov observe [1, §5.4.2] that the hypothesis actually needed is weaker — that the torus act on each V_i with *simple spectrum* — and ask for a combinatorial characterisation of such w. Each resulting poset $P = P(w)$ satisfies

$$\sum_{\varphi \in \mathrm{rpp}(P)} q^{|\varphi|} \;=\; \prod_{u \in P} \bigl(1 - q^{\,h(u)}\bigr)^{-1} \tag{1}$$

for some $h : P \to \mathbb{Z}_{>0}$, that is, is a *hook length poset* in the sense of [2]; the class strictly contains the d-complete posets. Nothing below depends on that general statement, since (1) is verified directly for the one poset we use.

The following was asked of the author by V. Krylov in October 2026.

Question 1. Naruse and Okada prove a *skew* hook formula [3, Thm. 1.2]: for d-complete P and an order ideal $F \subseteq P$, the series of reverse plane partitions on $P \setminus F$ is a finite sum over the excited diagrams of F in P, with hook function [3, Def. 2.6]. Does it extend to the skew-heaps $P \setminus F$ of the simple-spectrum setting, where P is a hook length poset but need not be d-complete?

Theorem 2. *No. There is an eight-element heap in D_4 which is a hook length poset, is not d-complete, and for which (1) fails for every hook function [3, Def. 2.6] produces. Hence [3, Thm. 1.2], read with its own hook function, already fails there at $F = \varnothing$.*

Only the degenerate case of the formula is needed. When $F = \varnothing$ the excited diagrams reduce to the empty one and the set of excited peaks is empty, so the formula becomes (1); this is the specialisation DKK quote at the end of their §5.3. So it suffices to rule out every admissible h, and the next lemma makes that a question about multisets, hence finite.

Lemma 3. *If $\prod_i (1 - q^{a_i})^{-1} = \prod_j (1 - q^{b_j})^{-1}$ in $\mathbb{Z}[[q]]$ for finite multisets of positive integers, the multisets coincide.*

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Proof. Let c_d be the multiplicity of d among the a_i. Then

$$\log \prod_i (1-q^{a_i})^{-1} = \sum_{m\geq 1} \frac{1}{m} \sum_i q^{a_i m} = \sum_{N\geq 1} q^N \sum_{d\mid N} c_d \, \frac{d}{N},$$

so $N[q^N]\log(\cdot) = \sum_{d\mid N} d\, c_d$ for all N. These values determine $d\, c_d$, hence c_d, by induction on N. □

We may therefore speak of the *true hook multiset* of a hook length poset: it is determined by the series alone.

2. The witness

Proposition 4. *Take $\Gamma = D_4$ in the Bourbaki labelling, so vertex 2 is trivalent with neighbours $1, 3, 4$, and let*

$$w = s_2 s_1 s_2 s_3 s_4 s_2 s_1 s_3,$$

reduced of length 8. Its heap P is

$$p_1 \prec p_2 \prec p_3 \prec \{p_4, p_5\} \prec p_6 \prec \{p_7, p_8\}, \qquad \text{colours } 2,1,2,3,4,2,1,3,$$

with p_4, p_5 incomparable (colours $3, 4$ non-adjacent) and likewise p_7, p_8 (colours $1, 3$). Then P is not d-complete, and its true hook multiset is $\{1,1,3,4,4,5,6,7\}$.

Proof. The order is the transitive closure of the heap relations read off the word. For d-completeness no computation is needed: p_7, p_8 are incomparable with common lower cover p_6, so they span a d_3^--interval, which Proctor's axiom at $k = 3$ requires to have a top; but both are maximal in P. The dual poset is not d-complete either, so this is independent of orientation.

For the multiset, both sides of (1) are computed independently and compared to order q^{26}. A reverse plane partition $\varphi : P \to \mathbb{Z}_{\geq 0}$, weakly increasing upwards, is equivalent to the decreasing chain of order filters $F_k = \{x : \varphi(x) \geq k\}$ with $|\varphi| = \sum_k |F_k|$; repetitions $F_k = F_{k+1}$ are allowed, so

$$\sum_{\varphi \in \mathrm{rpp}(P)} q^{|\varphi|} = 1 + \sum_{\varnothing \neq F} g(F), \qquad g(F) = \frac{q^{|F|}}{1-q^{|F|}} \Big(1 + \sum_{\varnothing \neq G \subsetneq F} g(G)\Big).$$

P has ten nonempty filters, and the recursion gives

$$1 + 2q + 3q^2 + 5q^3 + 9q^4 + 14q^5 + 21q^6 + 31q^7 + 45q^8 + 63q^9 + \cdots,$$

agreeing with $\prod_{h\in\{1,1,3,4,4,5,6,7\}} (1-q^h)^{-1}$ in every coefficient to q^{26}; Lemma 3 does the rest. The first two coefficients check by hand: $[q^1] = 2$ counts the maximal elements p_7, p_8, and $[q^2] = 3$ counts $2 \cdot \mathbf{1}_{p_7}$, $2 \cdot \mathbf{1}_{p_8}$ and $\mathbf{1}_{p_7} + \mathbf{1}_{p_8}$. □

Off the d-complete locus [3, Def. 2.6] does not define a single function. An element may top several d_k-intervals, and at each such element the recursion may use any one of them, or the order-ideal count $\#\{v : v \preceq u\}$ instead; so it defines a family. We exhaust it.

Proposition 5. *For P as above, three elements top a d_k-interval in the orientation of Proposition 4, giving eight choices and six distinct multisets,*

$$\{1,2,3,4,4,5,6,6\},\ \{1,2,3,4,4,5,6,7\},\ \{1,2,3,4,4,5,7,7\},$$
$$\{1,2,3,4,4,6,6,6\},\ \{1,2,3,4,4,6,6,7\},\ \{1,2,3,4,4,6,7,7\};$$

in the dual orientation — the corner-as-maximum convention of [3, Def. 2.6] *— two elements do, giving six choices and two multisets, $\{1,1,3,4,4,5,7,8\}$ and $\{1,1,3,4,4,6,7,8\}$. In neither case does $\{1,1,3,4,4,5,6,7\}$ occur.*

Proof. Finite enumeration, eight and six cases. □

Proof of Theorem 2. If [3, Thm. 1.2] held for P with hook function h from [3, Def. 2.6], then $F = \varnothing$ would give (1) for h, so by Lemma 3 the multiset of h would be the true hook multiset $\{1, 1, 3, 4, 4, 5, 6, 7\}$ of Proposition 4. Proposition 5 rules that out. □

Remark 6. Both orientations are recorded because the hook function is pinned where the series is a hook product while [3, Def. 2.6] is stated in the dual convention, and the conclusion should not rest on that correspondence. In the first orientation the failure is narrow, $\{1, 2, 3, 4, 4, 5, 6, 7\}$ differing in one entry; in the second the two largest hooks come out $7, 8$ against $6, 7$, so the product has the wrong order of vanishing at $q \to 1$.

3. What this does and does not rule out

What fails is [3, Thm. 1.2] *read with its own hook function*, on hook length heaps that are not d-complete, and it fails at $F = \varnothing$, before excited diagrams play any role. What does *not* fail is (1): the witness is a hook length poset, so the $F = \varnothing$ case of the formula is true for it with the hook function realising $\{1, 1, 3, 4, 4, 5, 6, 7\}$. Nothing here bears on d-complete P, where [3, Thm. 1.2] is a theorem, nor on any formula whose hook function is defined otherwise. So it is the recipe for h that is at fault, not the shape of the formula, and the natural question survives.

Question 7. Every simple-spectrum heap P carries a hook function h_P satisfying (1), unique as a multiset by Lemma 3. Is there a recursion computing h_P, agreeing with [3, Def. 2.6] on the d-complete locus, for which [3, Thm. 1.2] holds throughout this class?

For how typical the witness is: over all reduced words of length at most 8 in D_4, among the heaps that are hook length posets, the d-complete ones reach their true hook multiset in all 336 cases and the non-d-complete ones fail to in all 36. In this range the dichotomy is exactly d-completeness. All 36 failures have $|P| = 8$, so the witness is one of 36 of minimal size.

Every claim above is a finite check. The scripts are in the `skewheap` directory of https://github.com/jaideepsaipadhi/dkk-simple-spectrum: `certify2.py` computes both sides of (1) for the witness in either orientation, from scratch and sharing no code with the rest of the project; `orient.py` exhausts the [3, Def. 2.6] family and prints every reachable multiset; `minwit.py 8 D4` recovers the witness as a smallest one; `reach.py 8 11 D4` produces the 336 and 36.

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References

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Department of Computer Science, Purdue University, West Lafayette, IN 47907
Email address: jpadhi@purdue.edu